

1 What's Wrong with Maximum Likelihood?

Let $Y := (Y_1, \dots, Y_n)$ be a data set we observe and let $f(\cdot | \theta)$ be a probability density model where θ is the parameter. Suppose that we try and fit model $f(\cdot | \theta)$ with data Y to obtain the estimate of the parameter $\hat{\theta}_Y$ defined as follows:

$$\hat{\theta}_Y := \arg \max_{\theta} \log f(Y | \theta). \quad (\text{ML})$$

What are we actually doing here? We are supposing that *if* Y is drawn according to the density $f(\cdot | \theta_0)$ for some true parameter θ_0 , *then* $\hat{\theta}_Y$ is a good estimate for θ_0 . This line of reasoning is extensively argued by Ronald Fisher, the inventor of the **Maximum Likelihood (ML)** method.

Yet, this approach has an obvious problem: What if Y follows another distribution with density function $g(\cdot | \phi_0)$? We can, of course, also find the **ML** estimate for ϕ_0 :

$$\hat{\phi}_Y := \arg \max_{\phi} \log g(Y | \phi).$$

In the spirit of **ML**, a straightforward and naive approach is to see which model fits the data better, which means comparing the fitted log-likelihoods

$$\log f(Y | \hat{\theta}_Y) \quad \text{and} \quad \log g(Y | \hat{\phi}_Y) \quad (1)$$

and choose the model with the higher log-likelihood. This comparison is meaningful only if both models describe the same observations and their densities use the same dominating measure. This poses another problem: since we only have one realization of the data Y , we can always find some density function $h(\cdot | \psi)$ that is carefully tailored to fit the data at hand Y very well, producing a high fitted likelihood $h(Y | \hat{\psi}_Y)$. However, if another data set X drawn from the same distribution as Y is presented, the model $h(\cdot | \psi)$ may fit the new data X poorly. This is known as the problem of **overfitting**.

Luckily, in describing **overfitting**, we are motivated to do **cross-validation**, i.e., to use another data set X that is independent of Y but follows the same distribution to evaluate a parameter estimate obtained from Y .

2 Deriving **AIC**

Let's switch back to $f(\cdot | \theta)$, with fixed parameter dimension k as $n \rightarrow \infty$. For the derivation, let $X = (X_1, \dots, X_n)$ and $Y = (Y_1, \dots, Y_n)$ be independent samples from

[†]I assume the usual regularity conditions for maximum likelihood: an interior and identifiable true parameter, three times differentiable log-densities, nonsingular Fisher information, and sufficient moment conditions for the convergence of expectations below. I also assume the reader is familiar with the **ML** estimator, its asymptotic properties, and Fisher information.

$f(\cdot | \theta_0)$, and write

$$f(X | \theta) := \prod_{i=1}^n f(X_i | \theta)$$

for the joint density; define $f(Y | \theta)$ similarly. Instead of trying to estimate and compare the log-likelihoods in (1), we try to estimate their **cross-validated** versions

$$\log f(X | \hat{\theta}_Y) \quad \text{and} \quad \log g(X | \hat{\phi}_Y).$$

We derive the correction for f ; the same calculation applies to g under analogous correct-specification and regularity assumptions. Thus, $\hat{\theta}_Y$ is fitted on Y and evaluated on an independent data set X . Since X is unavailable, we instead approximate the expected **cross-validated** log-likelihood $\mathbf{E}_{X,Y}[\log f(X | \hat{\theta}_Y)]$.

First, expand $\log f(X | \hat{\theta}_Y)$ to second order around $\hat{\theta}_X$:

$$\begin{aligned} \log f(X | \hat{\theta}_Y) &= \log f(X | \hat{\theta}_X) && \text{(zeroth order)} \\ &+ (\hat{\theta}_Y - \hat{\theta}_X)' \left[\frac{\partial \log f(X | \hat{\theta}_X)}{\partial \theta} \right] && \text{(first order)} \\ &+ \frac{1}{2} (\hat{\theta}_Y - \hat{\theta}_X)' \left[\frac{\partial^2 \log f(X | \hat{\theta}_X)}{\partial \theta \partial \theta'} \right] (\hat{\theta}_Y - \hat{\theta}_X) + o_p(1). && \text{(second order)} \end{aligned}$$

The $o_p(1)$ remainder follows from the regularity conditions and $\hat{\theta}_Y - \hat{\theta}_X = O_p(n^{-1/2})$. The first-order term is exactly zero because $\hat{\theta}_X$ is an interior **ML** estimator. Thus,

$$\log f(X | \hat{\theta}_Y) = \log f(X | \hat{\theta}_X) + \frac{1}{2} (\hat{\theta}_Y - \hat{\theta}_X)' \mathbf{J}(\hat{\theta}_X) (\hat{\theta}_Y - \hat{\theta}_X) + o_p(1), \quad (2)$$

where

$$\mathbf{J}(\hat{\theta}_X) := \frac{\partial^2 \log f(X | \hat{\theta}_X)}{\partial \theta \partial \theta'}.$$

Here comes the key insight of **AIC**: the cross-validated log-likelihood $\log f(X | \hat{\theta}_Y)$ is equal, up to $o_p(1)$, to the fitted log-likelihood $\log f(X | \hat{\theta}_X)$ plus a correction term. The correction term involves the Hessian $\mathbf{J}(\hat{\theta}_X)$ and the difference between the two **ML** estimators $\hat{\theta}_Y - \hat{\theta}_X$. Now split the quadratic form into three parts:

$$(\hat{\theta}_Y - \hat{\theta}_X)' \mathbf{J}(\hat{\theta}_X) (\hat{\theta}_Y - \hat{\theta}_X) = (\hat{\theta}_X - \theta_0)' \mathbf{J}(\hat{\theta}_X) (\hat{\theta}_X - \theta_0) \quad (a)$$

$$+ (\hat{\theta}_Y - \theta_0)' \mathbf{J}(\hat{\theta}_X) (\hat{\theta}_Y - \theta_0) \quad (b)$$

$$- 2(\hat{\theta}_Y - \theta_0)' \mathbf{J}(\hat{\theta}_X) (\hat{\theta}_X - \theta_0). \quad (c)$$

We now take expectations. This point is essential: part (c) does *not* generally converge to zero in probability. Instead, the standard **ML** limits give

$$\begin{aligned} \sqrt{n}(\hat{\theta}_X - \theta_0) &\xrightarrow{d} \mathcal{N}_k(0, \mathcal{I}(\theta_0)^{-1}), & \text{and} & \quad \frac{\mathbf{J}(\hat{\theta}_X)}{n} \xrightarrow{p} -\mathcal{I}(\theta_0), \\ \sqrt{n}(\hat{\theta}_Y - \theta_0) &\xrightarrow{d} \mathcal{N}_k(0, \mathcal{I}(\theta_0)^{-1}), \end{aligned}$$

where the first two limiting normal vectors are independent and $\mathcal{I}(\theta_0)$ is the Fisher information for one observation. Their independence and zero means imply, under the stated moment conditions,

$$\mathbf{E}_{X,Y} \left[(\hat{\theta}_Y - \theta_0)' \mathbf{J}(\hat{\theta}_X) (\hat{\theta}_X - \theta_0) \right] \rightarrow 0. \quad (3)$$

Parts (a) and (b) are similar in form:

$$\begin{aligned} (\hat{\theta}_Y - \theta_0)' \mathbf{J}(\hat{\theta}_X) (\hat{\theta}_Y - \theta_0) &= \text{trace} \left(n(\hat{\theta}_Y - \theta_0)(\hat{\theta}_Y - \theta_0)' \frac{\mathbf{J}(\hat{\theta}_X)}{n} \right), \\ (\hat{\theta}_X - \theta_0)' \mathbf{J}(\hat{\theta}_X) (\hat{\theta}_X - \theta_0) &= \text{trace} \left(n(\hat{\theta}_X - \theta_0)(\hat{\theta}_X - \theta_0)' \frac{\mathbf{J}(\hat{\theta}_X)}{n} \right). \end{aligned}$$

Asymptotic normality, the information equality, and convergence of the relevant moments therefore give

$$\mathbf{E}_{X,Y}[\text{part (a)}] \rightarrow -k \quad \text{and} \quad \mathbf{E}_{X,Y}[\text{part (b)}] \rightarrow -k. \quad (4)$$

Thus it is the *expectation* of each part, rather than the random part itself, that is asymptotically $-k$. Taking expectations in (2) and using (3)–(4), we obtain

$$\mathbf{E}_{X,Y}[\log f(X | \hat{\theta}_Y)] = \mathbf{E}_X[\log f(X | \hat{\theta}_X)] - k + o(1). \quad (5)$$

Since X and Y have the same distribution, $\log f(Y | \hat{\theta}_Y) - k$ is therefore an asymptotically unbiased estimate of the expected cross-validated log-likelihood. This is the central correction behind **AIC**. It is conventionally written as

$$\text{AIC} := 2k - 2 \log f(Y | \hat{\theta}_Y). \quad (\text{AIC})$$

Its sign connects the criterion with information theory and **Kullback-Leibler Divergence**. The simple k correction uses correct specification and the information equality; for a misspecified family, the analogous correction need not equal k .

3 **AIC's Connection with Kullback-Leibler Divergence**

KL divergence is an information-theoretic measure of the discrepancy between two distributions. For densities p and q with respect to the same dominating measure, it is defined as

$$\text{KL}(p \parallel q) := \int_{\mathcal{X}} \log \left[\frac{p(x)}{q(x)} \right] p(x) dx,$$

provided the integral is well defined. The two main properties of **KL** are

1. $\text{KL}(p \parallel q) \geq 0$.
2. $\text{KL}(p \parallel q) = 0$ if and only if $p = q$ almost everywhere.

That is, $\text{KL}(p \parallel q)$ is small when p and q are similar.

In our case, we want to know the discrepancy between the “true” joint density $f(\cdot | \theta_0)$ and the fitted joint density $f(\cdot | \hat{\theta}_Y)$. Because the latter depends on the random training sample Y , the relevant discrepancy is its expectation:

$$\begin{aligned} & \mathbf{E}_Y [\text{KL}(f(\cdot | \theta_0) \parallel f(\cdot | \hat{\theta}_Y))] \\ &= \int_{\mathcal{X}} \log f(X | \theta_0) f(X | \theta_0) dX - \mathbf{E}_Y \left[\int_{\mathcal{X}} \log f(X | \hat{\theta}_Y) f(X | \theta_0) dX \right] \\ &= \text{constant} - \mathbf{E}_{X,Y} [\log f(X | \hat{\theta}_Y)]. \end{aligned}$$

Here $\mathcal{X}$ is the sample space of the full data set X . The first integral is the negative entropy of the true distribution and is constant across candidate models. The quantity $-\mathbf{E}_{X,Y} [\log f(X | \hat{\theta}_Y)]$ is the expected **cross-entropy**. Equation (5) therefore shows that

$$-2 \log f(Y | \hat{\theta}_Y) + 2k$$

is asymptotically unbiased for twice the expected cross-entropy. Thus minimizing expected cross-entropy is equivalent to minimizing expected **KL** divergence.

Measuring the discrepancy between $f(\cdot | \theta_0)$ and $f(\cdot | \hat{\theta}_Y)$ makes intuitive sense: the problem of **overfitting** can be understood as a large discrepancy between the “true” likelihood and the “estimated” likelihood. In the original paper (Akaike, 1974), **AIC** is motivated by **KL**. Hence, **AIC** is represented as the *negative* of the corrected log-likelihood to match the sign of **cross-entropy**. Thus, in practice, we select the model with the *smallest* **AIC**.

4 Why Times Two?

If we consider a Gaussian model with $\theta = (\mu, \sigma^2)$, the log-likelihood is written as

$$\log f(X | \theta) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2.$$

Multiplying the log-likelihood by two removes these factors of $\frac{1}{2}$. More generally, the factor of two is a conventional scale inherited from likelihood-ratio statistics and model deviance; it does not affect the ranking of candidate models. The minus sign turns maximization of corrected log-likelihood into minimization of **AIC**. ■

References

Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6), 716–723. <https://doi.org/10.1109/TAC.1974.1100705>